

Course 4281

4 Credits: 4

Programme Core Course

Source-grounded

Programming in Python

□ Provide knowledge in programming using Python language.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 4281 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 4281.pdf from the uploaded official SITTTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Cyber Forensics and Information security
Course code	4281
Course title	Programming in Python
Semester	4 Credits: 4
Course category	Programme Core Course
Periods per week	4 (L:4 T:0 P:0) Periods per semester: 60
Periods per semester	60

Item	Official information
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

41 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- Provide knowledge in programming using Python language.
- Discover the need of working with strings and functions.
- Illustrate the process of structuring the data using lists, dictionaries, tuples and sets.
- Familiarize exception handling and Object-Oriented Programming concepts.
- Design programs with file operations and database connectivity.

Course outcomes

CO	Official outcome	Study level
CO1	14 Applying semantics and control flow statements. Construct modular programs using functions and Understanding	See official syllabus
CO2	15 make use of strings handling functions. Applying Demonstrate exception handling and Object-Oriented Understanding Programming concepts such as encapsulation, 14	See official syllabus
CO3	Applying inheritance and polymorphism. Build programs using files and with database Understanding	See official syllabus
CO4	15 connectivity. Applying	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3 2
CO3	CO3 3 2

Row	Extracted official mapping
CO4	CO4 3 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Introduction to Python: Python variables, Python basic Operators, Understanding python	8	As specified
Module 2	Python Complex data types: Using string data type and string operations, Defining list	13	As specified
Module 3	Exceptions-Built-in Exceptions, Handling Exceptions, Exception with Arguments, User-	6	As specified
Module 4	Python File Operations: Reading files, Writing files in python, Understanding read functions,	14	As specified

MODULE 1

Introduction to Python: Python variables, Python basic Operators, Understanding python

This module is presented from the official syllabus scope for Programming in Python. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.

- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Introduction to Python: Python variables, Python basic Operators, Understanding python
- Official duration: As specified
- Official topic entries captured: 8
- The full source excerpt is preserved below for coverage checking.

blocks. Python Data Types, Declaring and using Numeric data types.

blocks. Python Data Types, Declaring and using Numeric data types. is an official topic in Module 1 of Programming in Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify blocks. Python Data Types, Declaring and using Numeric data types. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, blocks. python data types, declaring and using numeric data types. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What blocks. Python Data Types, Declaring and using Numeric data types. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1-☐ blocks. Python Data Types, Declaring and using Numeric data types. ☐ definition/meaning ☐. ☐ steps, application ☐. Technical terms English-☐.

Python Program Flow Control Conditional blocks: if, else and else if, Simple for loops in

Python Program Flow Control Conditional blocks: if, else and else if, Simple for loops in is an official topic in Module 1 of Programming in Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Python Program Flow Control Conditional blocks: if, else and else if, Simple for loops in using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, python program flow control conditional blocks: if, else and else if, simple for loops in should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Python Program Flow Control Conditional blocks: if, else and else if, Simple for loops in means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-പൈതൃക Python Program Flow Control Conditional blocks: if, else and else if, Simple for loops in പൈതൃക പൈതൃക പൈതൃക definition/meaning പൈതൃക പൈതൃക പൈതൃക, steps, application പൈതൃക പൈതൃക പൈതൃക. Technical terms English-പൈതൃക പൈതൃക പൈതൃക.

python, For loop using ranges, string, list and dictionaries. Use of while loops in python,

python, For loop using ranges, string, list and dictionaries. Use of while loops in python, is an official topic in Module 1 of Programming in Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify python, For loop using ranges, string, list and dictionaries. Use of while loops in python, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, python, for loop using ranges, string, list and dictionaries. use of while loops in python, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What python, For loop using ranges, string, list and dictionaries. Use of while loops in python, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-പൈതൃകം python, For loop using ranges, string, list and dictionaries. Use of while loops in python, പൈതൃകം definition/meaning പൈതൃകം. പൈതൃകം പൈതൃകം പൈതൃകം, steps, application പൈതൃകം പൈതൃകം പൈതൃകം. Technical terms English-പൈതൃകം പൈതൃകം പൈതൃകം.

Loop manipulation using pass, continue, break and else.

Loop manipulation using pass, continue, break and else. is an official topic in Module 1 of Programming in Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Loop manipulation using pass, continue, break and else. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, loop manipulation using pass, continue, break and else. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Loop manipulation using pass, continue, break and else. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-Loop manipulation using pass, continue, break and else.
Loop definition/meaning
Loop steps, application
Technical terms English-
Loop manipulation.

Programming using Python conditional and loop blocks.

Programming using Python conditional and loop blocks. is an official topic in Module 1 of Programming in Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Programming using Python conditional and loop blocks. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, programming using python conditional and loop blocks. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Programming using Python conditional and loop blocks. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Python Programming using Python conditional and loop blocks. Python definition/meaning Python steps, application Python. Technical terms English-Python.

functions.

functions. is an official topic in Module 1 of Programming in Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify functions. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, functions. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What functions. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

[illegible]

Understanding

Understanding is an official topic in Module 1 of Programming in Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understanding using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understanding should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understanding means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-Understanding definition/meaning . steps, application . Technical terms English-

Applying

Applying is an official topic in Module 1 of Programming in Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Applying using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, applying should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Applying means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-എപ്രയോഗിക്കുന്നതിനുള്ള നിർവ്വചന/അർത്ഥം എഴുതുക. എഴുതുക. എഴുതുക, steps, application എഴുതുക. Technical terms English-ൽ എഴുതുക.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Introduction to Python: Python variables, Python basic Operators, Understanding python blocks. Python Data Types, Declaring and using Numeric data types. Python Program Flow Control Conditional blocks: if, else and else if, Simple for loops in python, For loop using ranges, string, list and dictionaries. Use of while loops in python, Loop manipulation using pass, continue, break and else. Programming using Python conditional and loop blocks.

Construct modular programs using functions and make use of strings handling

C02

functions.

M2.01	Explain Complex data types	2
Understanding		
M2.02	Demonstrate String manipulation methods	4
Understanding		
M2.03	Demonstrate List manipulation	3
Understanding		
Understanding		
M2.04	Build programs using functions.	5
Applying		

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

Python Complex data types: Using string data type and string operations, Defining list

This module is presented from the official syllabus scope for Programming in Python. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Python Complex data types: Using string data type and string operations, Defining list
- Official duration: As specified
- Official topic entries captured: 13
- The full source excerpt is preserved below for coverage checking.

Python Complex data types: Using string data type and string operations, Defining list

Python Complex data types: Using string data type and string operations, Defining list is an official topic in Module 2 of Programming in Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Python Complex data types: Using string data type and string operations, Defining list using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, python complex data types: using string data type and string operations, defining list should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Python Complex data types: Using string data type and string operations, Defining list means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-പിതൃ Python Complex data types: Using string data type and string operations, Defining list പട്ടിക definition/meaning പട്ടിക. പട്ടിക പട്ടിക പട്ടിക, steps, application പട്ടിക പട്ടിക പട്ടിക. Technical terms English-പട്ടിക പട്ടിക പട്ടിക.

and list slicing, Use of Tuple data type. String, List and Dictionary manipulations -

and list slicing, Use of Tuple data type. String, List and Dictionary manipulations - is an official topic in Module 2 of Programming in Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify and list slicing, Use of Tuple data type. String, List and Dictionary manipulations - using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, and list slicing, use of tuple data type. string, list and dictionary manipulations - should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What and list slicing, Use of Tuple data type. String, List and Dictionary manipulations - means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- π and list slicing, Use of Tuple data type. String, List and Dictionary manipulations - π definition/meaning π . π steps, application π . Technical terms English- π .

string manipulation methods, List manipulation, Dictionary manipulation, Programming

string manipulation methods, List manipulation, Dictionary manipulation, Programming is an official topic in Module 2 of Programming in Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify string manipulation methods, List manipulation, Dictionary manipulation, Programming using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, string manipulation methods, list manipulation, dictionary manipulation, programming should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What string manipulation methods, List manipulation, Dictionary manipulation, Programming means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-
string manipulation methods, List manipulation, Dictionary manipulation, Programming definition/meaning . steps, application . Technical terms English-

using string, list and dictionary, Python Functions - in-built and user defined functions,

using string, list and dictionary, Python Functions - in-built and user defined functions, is an official topic in Module 2 of Programming in Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify using string, list and dictionary, Python Functions - in-built and user defined functions, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, using string, list and dictionary, python functions - in-built and user defined functions, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What using string, list and dictionary, Python Functions - in-built and user defined functions, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-പുറം using string, list and dictionary, Python Functions - in-built and user defined functions, പദപരിവർത്തനം പദപരിവർത്തനം definition/meaning പദപരിവർത്തനം. പദപരിവർത്തനം പദപരിവർത്തനം, steps, application പദപരിവർത്തനം പദപരിവർത്തനം പദപരിവർത്തനം. Technical terms English-പദ പദപരിവർത്തനം പദപരിവർത്തനം.

Organizing python codes using functions.

Organizing python codes using functions. is an official topic in Module 2 of Programming in Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Organizing python codes using functions. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, organizing python codes using functions. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Organizing python codes using functions. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ഓർഗനൈസിംഗ് പൈതൻ കോഡ് ഉപയോഗിച്ച് ഫങ്ഷനുകൾ.

ഓർഗനൈസിംഗ് പൈതൻ കോഡ് definition/meaning ഓർഗനൈസിംഗ് പൈതൻ കോഡ്. ഓർഗനൈസിംഗ് പൈതൻ കോഡ്, steps, application ഓർഗനൈസിംഗ് പൈതൻ കോഡ്. Technical terms English-ഓർഗനൈസിംഗ് പൈതൻ കോഡ്.

Demonstrate exception handling and Object-Oriented Programming concepts

Demonstrate exception handling and Object-Oriented Programming concepts is an official topic in Module 2 of Programming in Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Demonstrate exception handling and Object-Oriented Programming concepts using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, demonstrate exception handling and object-oriented programming concepts should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Demonstrate exception handling and Object-Oriented Programming concepts means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 2- Demonstrate exception handling and Object-Oriented Programming concepts definition/meaning . . . , steps, application Technical terms English-

such as encapsulation, inheritance and polymorphism.

such as encapsulation, inheritance and polymorphism. is an official topic in Module 2 of Programming in Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify such as encapsulation, inheritance and polymorphism. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, such as encapsulation, inheritance and polymorphism. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What such as encapsulation, inheritance and polymorphism. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ഉൾപ്പെടെ encapsulation, inheritance and polymorphism. ഉപയോഗിക്കുന്നതിന് definition/meaning ഉപയോഗിക്കുക. ഉപയോഗിക്കുന്നതിന് ഉപയോഗിക്കുക, steps, application ഉപയോഗിക്കുക ഉപയോഗിക്കുക ഉപയോഗിക്കുക. Technical terms English-ൽ ഉപയോഗിക്കുക ഉപയോഗിക്കുക.

Illustrate Exception with Arguments, User-

Illustrate Exception with Arguments, User- is an official topic in Module 2 of Programming in Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate Exception with Arguments, User- using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate exception with arguments, user- should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate Exception with Arguments, User- means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ഈ Illustrate Exception with Arguments, User-
ഈ definition/meaning ഈ. ഈ ഈ, steps, application ഈ ഈ. Technical terms English-ഈ ഈ ഈ.

defined Exceptions

defined Exceptions is an official topic in Module 2 of Programming in Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify defined Exceptions using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, defined exceptions should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What defined Exceptions means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-എന്നിവിടെ നിർവ്വചിച്ചിരിക്കുന്ന വ്യതിയാനങ്ങൾ (Exceptions) നിർവ്വചനം/അർത്ഥം (definition/meaning) ഉൾക്കൊള്ളുന്നു. വ്യതിയാനങ്ങൾ (Exceptions) നിർവ്വചിക്കുന്നതിനുള്ള ഓരോ ഓരോ ഓരോ ഓരോ, steps, application ഉൾക്കൊള്ളുന്നു. Technical terms English-ൽ ഉൾക്കൊള്ളുന്നു.

Understanding

Understanding is an official topic in Module 2 of Programming in Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understanding using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understanding should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understanding means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ഉറപ്പ് Understanding പദപരിവർത്തനം പദപരിവർത്തനം definition/meaning പദപരിവർത്തനം. പദപരിവർത്തനം പദപരിവർത്തനം, steps, application പദപരിവർത്തനം പദപരിവർത്തനം പദപരിവർത്തനം. Technical terms English-ഉറപ്പ് പദപരിവർത്തനം പദപരിവർത്തനം.

Applying

Applying is an official topic in Module 2 of Programming in Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Applying using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, applying should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Applying means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-എന്നുപയോഗിക്കുന്ന പദങ്ങളുടെ
definition/meaning നൽകുക. പദങ്ങൾ പദങ്ങൾ, steps, application
പദങ്ങൾ പദങ്ങൾ നൽകുക. Technical terms English-എന്ന പദങ്ങൾ നൽകുക.

Demonstrate Encapsulation, Inheritance and Understanding

Demonstrate Encapsulation, Inheritance and Understanding is an official topic in Module 2 of Programming in Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Demonstrate Encapsulation, Inheritance and Understanding using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, demonstrate encapsulation, inheritance and understanding should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Demonstrate Encapsulation, Inheritance and Understanding means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 2- Demonstrate Encapsulation, Inheritance and Understanding definition/meaning . steps, application . Technical terms English- .

Polymorphism. Applying

Polymorphism. Applying is an official topic in Module 2 of Programming in Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Polymorphism. Applying using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, polymorphism. applying should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Polymorphism. Applying means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-പോളിമോർഫിസം. Applying പദപരിച്ഛേദനം പദപരിച്ഛേദനം definition/meaning പദപരിച്ഛേദനം. പദപരിച്ഛേദനം പദപരിച്ഛേദനം, steps, application പദപരിച്ഛേദനം പദപരിച്ഛേദനം പദപരിച്ഛേദനം. Technical terms English-പോളിമോർഫിസം പദപരിച്ഛേദനം.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Python Complex data types: Using string data type and string operations, Defining list and list slicing, Use of Tuple data type. String, List and Dictionary manipulations - string manipulation methods, List manipulation, Dictionary manipulation, Programming using string, list and dictionary, Python Functions - in-built and user defined functions, Organizing python codes using functions.

concepts
C03 Demonstrate exception handling and Object-Oriented Programming such as encapsulation, inheritance and polymorphism.

M3.01 Understand Exception handling 1

M3.01 Understand Illustrate Exception with Arguments, User-defined Exceptions 2

M3.02 Understand Explain Object-Oriented Programming. 2

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

Exceptions-Built-in Exceptions, Handling Exceptions, Exception with Arguments, User-

This module is presented from the official syllabus scope for Programming in Python. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Exceptions-Built-in Exceptions, Handling Exceptions, Exception with Arguments, User-
- Official duration: As specified
- Official topic entries captured: 6
- The full source excerpt is preserved below for coverage checking.

Exceptions-Built-in Exceptions, Handling Exceptions, Exception with Arguments, User-

Exceptions-Built-in Exceptions, Handling Exceptions, Exception with Arguments, User- is an official topic in Module 3 of Programming in Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Exceptions-Built-in Exceptions, Handling Exceptions, Exception with Arguments, User- using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, exceptions-built-in exceptions, handling exceptions, exception with arguments, user- should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Exceptions-Built-in Exceptions, Handling Exceptions, Exception with Arguments, User- means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-എക്സെപ്ഷനുകൾ-Built-in Exceptions, Handling Exceptions, Exception with Arguments, User- ഡിഫൈൻഡ് എക്സെപ്ഷനുകൾ definition/meaning എക്സെപ്ഷനുകൾ. എക്സെപ്ഷനുകൾ എക്സെപ്ഷനുകൾ, steps, application എക്സെപ്ഷനുകൾ എക്സെപ്ഷനുകൾ എക്സെപ്ഷനുകൾ. Technical terms English-എക്സെപ്ഷനുകൾ എക്സെപ്ഷനുകൾ.

defined Exceptions.

defined Exceptions. is an official topic in Module 3 of Programming in Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify defined Exceptions. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, defined exceptions. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What defined Exceptions. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- defined Exceptions. definition/meaning . steps, application . Technical terms English-

Objects in Python, Constructor Method, Classes with Multiple Objects, Class Attributes

Objects in Python, Constructor Method, Classes with Multiple Objects, Class Attributes is an official topic in Module 3 of Programming in Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Objects in Python, Constructor Method, Classes with Multiple Objects, Class Attributes using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, objects in python, constructor method, classes with multiple objects, class attributes should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Objects in Python, Constructor Method, Classes with Multiple Objects, Class Attributes means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ഓബ്ജക്റ്റുകൾ Python-ൽ, Constructor Method, Classes with Multiple Objects, Class Attributes ഓബ്ജക്റ്റുകളുടെ definition/meaning എഴുതുക. ഓബ്ജക്റ്റുകൾ എങ്ങനെ ഉണ്ടാക്കുന്നു, steps, application എഴുതുക. Technical terms English-ൽ എഴുതുക.

versus Data Attributes, Encapsulation, Inheritance, Polymorphism.

versus Data Attributes, Encapsulation, Inheritance, Polymorphism. is an official topic in Module 3 of Programming in Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify versus Data Attributes, Encapsulation, Inheritance, Polymorphism. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, versus data attributes, encapsulation, inheritance, polymorphism. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What versus Data Attributes, Encapsulation, Inheritance, Polymorphism. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 3- versus Data Attributes, Encapsulation, Inheritance, Polymorphism. definition/meaning application, steps, application . Technical terms English-

Understanding

Understanding is an official topic in Module 3 of Programming in Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understanding using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understanding should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understanding means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-Understanding പദപരിചയം പദപരിചയം definition/meaning പദപരിചയം. പദപരിചയം പദപരിചയം, steps, application പദപരിചയം പദപരിചയം പദപരിചയം. Technical terms English-പദപരിചയം പദപരിചയം.

Applying

Applying is an official topic in Module 3 of Programming in Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Applying using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, applying should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Applying means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-എപ്ലൈംഗ് ഡിഫിനിഷൻ/മീനിംഗ് ഡിഫിനിഷൻ/മീനിംഗ്. ഡിഫിനിഷൻ ഡിഫിനിഷൻ, steps, application ഡിഫിനിഷൻ ഡിഫിനിഷൻ. Technical terms English-എ ഡിഫിനിഷൻ ഡിഫിനിഷൻ.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Exceptions-Built-in Exceptions, Handling Exceptions, Exception with Arguments, User-defined Exceptions.

Object-Oriented Programming, Classes and Objects, Creating Classes in Python, Creating Objects in Python, Constructor Method, Classes with Multiple Objects, Class Attributes versus Data Attributes, Encapsulation, Inheritance, Polymorphism.

C04 Build programs using files and with database connectivity.

M4.01	Demonstrate Python File Operations	3
Understanding		

M4.02	Understand read/write functions	3
Understanding		

Understanding		
M4.03	Utilize file pointers	3

Applying

Understanding		
M4.04	Develop Database Programming	5

Applying

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

Python File Operations: Reading files, Writing files in python, Understanding read functions,

This module is presented from the official syllabus scope for Programming in Python. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Python File Operations: Reading files, Writing files in python, Understanding read functions,
- Official duration: As specified
- Official topic entries captured: 14
- The full source excerpt is preserved below for coverage checking.

Manipulating file pointer using seek Programming, using file operations.: Connecting to a

Manipulating file pointer using seek Programming, using file operations.: Connecting to a is an official topic in Module 4 of Programming in Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Manipulating file pointer using seek Programming, using file operations.: Connecting to a using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, manipulating file pointer using seek programming, using file operations.: connecting to a should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Manipulating file pointer using seek Programming, using file operations.: Connecting to a means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഓപ്പറേറ്റിംഗ് മാനുവൽ Manipulating file pointer using seek Programming, using file operations.: Connecting to a ഓപ്പറേറ്റിംഗ് മാനുവൽ definition/meaning ഓപ്പറേറ്റിംഗ് മാനുവൽ. ഓപ്പറേറ്റിംഗ് മാനുവൽ ഓപ്പറേറ്റിംഗ്, steps, application ഓപ്പറേറ്റിംഗ് മാനുവൽ ഓപ്പറേറ്റിംഗ്. Technical terms English-ഓപ്പറേറ്റിംഗ് മാനുവൽ ഓപ്പറേറ്റിംഗ്.

Text / References

Text / References is an official topic in Module 4 of Programming in Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Text / References using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, text / references should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Text / References means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-Text / References ഓരോന്നിനും നിർവ്വചനം/അർത്ഥം നൽകുക. ഓരോന്നിന്റെയും ഓരോന്നിന്റെയും, steps, application നൽകുക. Technical terms English-ൽ നൽകുക.

T/R Book Title/Author

T/R Book Title/Author is an official topic in Module 4 of Programming in Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify T/R Book Title/Author using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, t/r book title/author should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What T/R Book Title/Author means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-
T/R Book Title/Author
definition/meaning .
 . steps, application
 . Technical terms English-
 .

Downey, A. et al., "How to think like a Computer Scientist: Learning with Python",

Downey, A. et al., "How to think like a Computer Scientist: Learning with Python", is an official topic in Module 4 of Programming in Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Downey, A. et al., "How to think like a Computer Scientist: Learning with Python", using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, downey, a. et al., "how to think like a computer scientist: learning with python", should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Downey, A. et al., "How to think like a Computer Scientist: Learning with Python", means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഡൗനിയുടെ, A. et al., "How to think like a Computer Scientist: Learning with Python", ഡിഫിനിഷൻ/മനസ്സിലാക്കൽ definition/meaning ഡിഫിനിഷൻ/മനസ്സിലാക്കൽ. ഡിഫിനിഷൻ/മനസ്സിലാക്കൽ, steps, application ഡിഫിനിഷൻ/മനസ്സിലാക്കൽ ഡിഫിനിഷൻ/മനസ്സിലാക്കൽ. Technical terms English-ൽ ഡിഫിനിഷൻ/മനസ്സിലാക്കൽ.

John Wiley, 2015 is an official topic in Module 4 of Programming in Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify John Wiley, 2015 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, John Wiley, 2015 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What John Wiley, 2015 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- John Wiley, 2015 ഐക്യരാഷ്ട്രസഭയുടെ നിയമം
definition/meaning ഐക്യരാഷ്ട്രസഭയുടെ നിയമം. ഐക്യരാഷ്ട്രസഭയുടെ നിയമം, steps, application
ഐക്യരാഷ്ട്രസഭയുടെ നിയമം. Technical terms English- ഐക്യരാഷ്ട്രസഭയുടെ നിയമം.

Jeeva Jose &P.SojanLal, "Introduction to Computing and Problem Solving with is an official topic in Module 4 of Programming in Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Jeeva Jose &P.SojanLal, "Introduction to Computing and Problem Solving with using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, jeeva jose &p.sojanlal, "introduction to computing and problem solving with should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Jeeva Jose &P.SojanLal, "Introduction to Computing and Problem Solving with means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 4-ജീവ ജോസെ & P. Sojan Lal, "Introduction to Computing and Problem Solving with പരിഹാരം definition/meaning പരിഹാരം. പരിഹാരം പരിഹാരം പരിഹാരം, steps, application പരിഹാരം പരിഹാരം പരിഹാരം. Technical terms English-പരിഹാരം പരിഹാരം പരിഹാരം.

PYTHON", Khanna Publishers, New Delhi, 2016 is an official topic in Module 4 of Programming in Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify PYTHON", Khanna Publishers, New Delhi, 2016 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, python", khanna publishers, new delhi, 2016 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What PYTHON", Khanna Publishers, New Delhi, 2016 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-“ PYTHON”, Khanna Publishers, New Delhi, 2016
പ്രോഗ്രാമിംഗ് ഭാഷയുടെ definition/meaning വിശദീകരിക്കുക. പ്രോഗ്രാമിംഗ് ഭാഷയുടെ പ്രയോഗം,
steps, application വിശദീകരിക്കുക. Technical terms English-ൽ വിശദീകരിക്കുക.

Wesley J. Chun, “Core Python Applications Programming”, 3rd Edition , Pearson is an official topic in Module 4 of Programming in Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Wesley J. Chun, “Core Python Applications Programming”, 3rd Edition , Pearson using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, wesley j. chun, “core python applications programming”, 3rd edition , pearson should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Wesley J. Chun, “Core Python Applications Programming”, 3rd Edition , Pearson means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-
Wesley J. Chun, “Core Python Applications Programming”, 3rd Edition , Pearson
 definition/meaning
 , steps, application
 . Technical terms English-
 .

Education, 2016

Education, 2016 is an official topic in Module 4 of Programming in Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Education, 2016 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, education, 2016 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Education, 2016 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-എന്ന Education, 2016 പാഠ്യപുസ്തകത്തിൽ നിന്ന്
definition/meaning ഉൾപ്പെടുത്തിയിരിക്കുന്നു. പാഠ്യപുസ്തകത്തിൽ നിന്ന്, steps, application
ഉൾപ്പെടുത്തിയിരിക്കുന്നു. Technical terms English-ൽ ഉൾപ്പെടുത്തിയിരിക്കുന്നു.

R2 Charles Dierbach, “Introduction to Computer Science using Python”, Wiley, 2015

R2 Charles Dierbach, “Introduction to Computer Science using Python”, Wiley, 2015 is an official topic in Module 4 of Programming in Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify R2 Charles Dierbach, “Introduction to Computer Science using Python”, Wiley, 2015 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, r2 charles dierbach, “introduction to computer science using python”, wiley, 2015 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What R2 Charles Dierbach, “Introduction to Computer Science using Python”, Wiley, 2015 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-പിറ R2 Charles Dierbach, “Introduction to Computer Science using Python”, Wiley, 2015 പദപരിചയം definition/meaning പദപരിചയം. പദപരിചയം പദപരിചയം, steps, application പദപരിചയം പദപരിചയം പദപരിചയം. Technical terms English-പദ പദപരിചയം പദപരിചയം.

Online Resources

Online Resources is an official topic in Module 4 of Programming in Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Online Resources using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, online resources should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Online Resources means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-
definition/meaning . , steps, application
Technical terms English-

SI No Website Link

SI No Website Link is an official topic in Module 4 of Programming in Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify SI No Website Link using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, SI no website link should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What SI No Website Link means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[SI No Website Link](#) [പ്രാഥമിക സാങ്കേതിക വിദ്യ](#)
definition/meaning [പ്രാഥമിക സാങ്കേതിക വിദ്യ](#). [പ്രാഥമിക സാങ്കേതിക വിദ്യ](#), steps, application
[പ്രാഥമിക സാങ്കേതിക വിദ്യ](#) [പ്രാഥമിക സാങ്കേതിക വിദ്യ](#). Technical terms English-[പ്രാഥമിക സാങ്കേതിക വിദ്യ](#).

1 <http://coursera.org/course/pythonlearn>

1 <http://coursera.org/course/pythonlearn> is an official topic in Module 4 of Programming in Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 <http://coursera.org/course/pythonlearn> using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 <http://coursera.org/course/pythonlearn> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 http://coursera.org/course/pythonlearn means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-പാഠ 1 <http://coursera.org/course/pythonlearn>
പാഠത്തിന്റെ definition/meaning പരിശോധിക്കുക. പാഠത്തിന്റെ പാഠങ്ങൾ,
steps, application പാഠത്തിൽ പരിശോധിക്കുക. Technical terms English-ൽ പാഠത്തിൽ
പരിശോധിക്കുക.

2 <http://spoken-tutorial.org/>

2 <http://spoken-tutorial.org/> is an official topic in Module 4 of Programming in Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 <http://spoken-tutorial.org/> using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 <http://spoken-tutorial.org/> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 http://spoken-tutorial.org/ means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-പാഠ്യ 2 <http://spoken-tutorial.org/> പാഠ്യപരിഭാഷണം
പാഠ്യ definition/meaning പാഠ്യപരിഭാഷണം. പാഠ്യപരിഭാഷണം പാഠ്യപരിഭാഷണം, steps,
application പാഠ്യപരിഭാഷണം പാഠ്യപരിഭാഷണം. Technical terms English-പാഠ്യ
പാഠ്യപരിഭാഷണം.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Python File Operations: Reading files, Writing files in python, Understanding
read functions,
read(), readline(), readlines(). Understanding write functions, write() and
writelines()
Manipulating file pointer using seek Programming, using file operations.:
Connecting to a
database, Creating Tables, INSERT, UPDATE, DELETE and READ operations, Transaction
Control, Exception Handling in Databases.

Text / References

T/R	Book Title/Author
Downey, A. et al., "How to think like a Computer Scientist: Learning with Python", T1	John Wiley, 2015
Jeeva Jose & P. Sojan Lal, "Introduction to Computing and Problem Solving with T2	PYTHON", Khanna Publishers, New Delhi, 2016
Wesley J. Chun, "Core Python Applications Programming", 3rd Edition , Pearson R1	Education, 2016
R2	Charles Dierbach, "Introduction to Computer Science using Python", Wiley, 2015

Module study routine: Read the source wording, make a one-page concept map,
practise one short answer and one structured answer, then review the Malayalam
support notes.

Concept and formula bank

Answer structure

Definition → Principle → Components/steps → Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure → Observation → Result → Safety

Use this sequence for laboratory, workshop and field activities.

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain blocks. Python Data Types, Declaring and using Numeric data types.. (3 marks)

Answer: Begin with the standard definition or identity of *blocks*. *Python Data Types, Declaring and using Numeric data types..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Python Program Flow Control Conditional blocks: if, else and else if, Simple for loops in. (3 marks)

Answer: Begin with the standard definition or identity of *Python Program Flow Control Conditional blocks: if, else and else if, Simple for loops in*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain python, For loop using ranges, string, list and dictionaries. Use of while loops in python,. (3 marks)

Answer: Begin with the standard definition or identity of *python, For loop using ranges, string, list and dictionaries. Use of while loops in python,.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Loop manipulation using pass, continue, break and else.. (3 marks)

Answer: Begin with the standard definition or identity of *Loop manipulation using pass, continue, break and else..* State the governing principle, list the key components or

stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 2

Define or explain Python Complex data types: Using string data type and string operations, Defining list. (3 marks)

Answer: Begin with the standard definition or identity of *Python Complex data types: Using string data type and string operations, Defining list*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain and list slicing, Use of Tuple data type. String, List and Dictionary manipulations -. (3 marks)

Answer: Begin with the standard definition or identity of *and list slicing, Use of Tuple data type. String, List and Dictionary manipulations -*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain string manipulation methods, List manipulation, Dictionary manipulation, Programming. (3 marks)

Answer: Begin with the standard definition or identity of *string manipulation methods, List manipulation, Dictionary manipulation, Programming*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain using string, list and dictionary, Python Functions - in-built and user defined functions,. (3 marks)

Answer: Begin with the standard definition or identity of *using string, list and dictionary, Python Functions - in-built and user defined functions,*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 3

Define or explain Exceptions-Built-in Exceptions, Handling Exceptions, Exception with Arguments, User-. (3 marks)

Answer: Begin with the standard definition or identity of *Exceptions-Built-in Exceptions, Handling Exceptions, Exception with Arguments, User-*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain defined Exceptions.. (3 marks)

Answer: Begin with the standard definition or identity of *defined Exceptions..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Objects in Python, Constructor Method, Classes with Multiple Objects, Class Attributes. (3 marks)

Answer: Begin with the standard definition or identity of *Objects in Python, Constructor Method, Classes with Multiple Objects, Class Attributes*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain versus Data Attributes, Encapsulation, Inheritance, Polymorphism.. (3 marks)

Answer: Begin with the standard definition or identity of *versus Data Attributes, Encapsulation, Inheritance, Polymorphism..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 4

Define or explain Manipulating file pointer using seek Programming, using file operations.: Connecting to a. (3 marks)

Answer: Begin with the standard definition or identity of *Manipulating file pointer using seek Programming, using file operations.: Connecting to a*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps,

□□□□□ application □□□□ □□□□□□□□ □□□□□.

Define or explain Text / References. (3 marks)

Answer: Begin with the standard definition or identity of *Text / References*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□ principle/steps, □□□□□ application □□□□ □□□□□□□□ □□□□□.

Define or explain T/R Book Title/Author. (3 marks)

Answer: Begin with the standard definition or identity of *T/R Book Title/Author*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□ principle/steps, □□□□□ application □□□□ □□□□□□□□ □□□□□.

Define or explain Downey, A. et al., "How to think like a Computer Scientist: Learning with Python",. (3 marks)

Answer: Begin with the standard definition or identity of *Downey, A. et al., "How to think like a Computer Scientist: Learning with Python",.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□ principle/steps, □□□□□ application □□□□ □□□□□□□□ □□□□□.

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **blocks. Python Data Types, Declaring and using Numeric data types..**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Python Complex data types: Using string data type and string operations, Defining list.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Exceptions-Built-in Exceptions, Handling Exceptions, Exception with Arguments, User-.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Manipulating file pointer using seek**
Programming, using file operations.: Connecting to a.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.

Term	Meaning in this lesson
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

- T/R Book Title/Author
- Downey, A. et al., "How to think like a Computer Scientist: Learning with Python", John Wiley, 2015
- Jeeva Jose & P. Sojan Lal, "Introduction to Computing and Problem Solving with PYTHON", Khanna Publishers, New Delhi, 2016
- Wesley J. Chun, "Core Python Applications Programming", 3rd Edition, Pearson Education, 2016
- R2 Charles Dierbach, "Introduction to Computer Science using Python", Wiley, 2015
- Online Resources
- SI No Website Link <http://coursera.org/course/pythonlearn> <http://spoken-tutorial.org/>

Web resources listed in the official PDF

- SI No Website Link <http://coursera.org/course/pythonlearn> <http://spoken-tutorial.org/>

IN-PAGE SEARCH

Search this lesson

Prepared from the supplied official SITTTR syllabus PDF for Course 4281. Verify institutional updates against the current official curriculum.